

A Hot Bath: Conserving Water Without Losing Uniformity

Summary

Maintaining a warm bath is a coupled problem of heat loss, replenishment and transport. A hot inlet can improve the mean temperature while leaving distant water cool and sending useful heat directly to the overflow. We therefore minimize added water subject to a lower limit in every cell and upper-temperature and spread limits outside a 0.15 m inlet zone, rather than optimizing an average alone.

We prove a coast-then-hold policy optimal for a well-mixed bath, then use a three-dimensional finite-volume thermal network with coefficients derived from textbook correlations and tied to a published immersion study where one exists (Section 2). The best constant rate is compared with an optimized piecewise-constant schedule; neither is proved optimal among all controls.

In a 164.25 L water-volume scenario lasting 30 minutes, with an initial temperature of 40°C and a 39°C lower limit, the best accepted constant-rate candidate adds **24.14 L** at 0.805 L/min from the start, and a 6-segment schedule with a 0.1°C design margin needs **21.48 L**, 11% less. The well-mixed optimum is 16.01 L, while an independent energy argument gives a 15.41 L lower bound for the spatial problem. A finer mesh changes the constant-rate result by 0.11%, and the selected scheduled policy passes independent continuous-time bounds on all three tested meshes.

Across 64 Sobol draws over the stated parameter ranges, 41 yield an accepted constant-rate candidate, needing 12–31 L (5th–95th percentile). Weak mixing often defeats this finite search. Motion improves transport, but the associated extra surface loss can offset the saving. The assumed foam layer reduces constant-flow demand to 10.4 L; this is a scenario, not a measured additive effect.

Geometry and mixing change the recommendation: a deep, narrow tub needs 17.48 L at equal volume, and weaker mixing moves inflow to the start. Cooling cannot identify the inlet route, and alternative flow/body mechanisms fit two probes. Passive/pulse trials retain 1465/467 pairs; at common reserves, candidates use 26.09/23.63 L across 5796 model-grid envelopes. A reusable 6 L pulse trial pays from unchanged use 3 with equal resets. Probe updates reduce the passive backup to 23.59/23.16 L in two nominal noise cases. Two off-grid realized schedules also satisfy physical limits in narrow four-parameter neighborhoods; changed physics can invalidate reuse. These conditional results do not establish safe fault recovery.

Keywords: thermal network; energy balance; lower bound; optimal control; sensitivity analysis

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1 Define the decision before optimizing

The task is to preserve both warmth and spatial uniformity in an overflowing, unheated tub, and to examine geometry, the bather and motion, and a bubble-bath layer [1]. The report separates physical requirements from preference assumptions. There is no supplied temperature record or measured heat-transfer coefficient to fit.

The decision variables are an inlet flow rate and the time at which a constant trickle begins. The tub is already full: added water displaces an equal volume through the overflow. The horizon is 1,800 s. Our baseline requires every cell average to stay at or above 39°C, and requires the cell averages outside a 0.15 m jet-mixing zone around the inlet to stay at or below 41°C and within an instantaneous spread of 1.5°C. These choices operationalize comfort; they are not medical limits or numbers specified by the problem, and the jet zone is an assumption justified in Section 2.1.

$$J = 1000 \int_0^{t_f} q(t) dt \quad (1)$$

$$T_i(t) \geq T_{\min} \quad (\forall i), \quad T_i(t) \leq T_{\max}, \quad \max_{i \in \Omega} T_i(t) - \min_{i \in \Omega} T_i(t) \leq \Delta \quad (i \in \Omega) \quad (2)$$

Here Ω is the set of cells outside the inlet jet zone, q is in m³/s and J is in litres. We prioritize the least water within the constraints, rather than assigning arbitrary weights to unlike units. Tightening the temperature tolerance is a separate scenario. Zero flow is admitted: if the initial stored heat suffices, using no added water is globally water-minimal.

Table 1 maps each requirement to its answer.

Table 1. Where each requirement is answered

Requirement	Where answered
Temperature in space and time	Sections 3, 5–7; Figures 1–2
Water-use strategy and its scope	Sections 4, 6–7 and 12
Tub/body geometry, size and temperature	Section 8
Motion and bubble-bath additive	Section 9
Validation and sensitivity	Sections 8–11
One-page non-technical explanation	Section 13

Spatial temperature means a control-volume average. It does not bound the unresolved temperature of a faucet jet or a skin-contact film. This distinction determines which practical conclusions the simulation can support.

Modeling options. A lumped model (Newton cooling) keeps one temperature and cannot see a cold corner. A flow solver resolves velocity but needs a turbulence closure and a faucet jet that nothing available here can constrain. A finite-volume thermal network sits between them: it keeps conservation exact, resolves where heat is lost and delivered, and makes mixing an explicit, testable parameter [5]. We use the lumped model as a proved benchmark (Sections 3–4) and the network for the decision (Sections 5–11).

Table 2 contrasts spatial resolution and data requirements.

Table 2. Modeling options compared

Option	Resolves	Evidence it needs	Role here
Lumped (Newton)	Mean temperature	Two loss coefficients	Proved benchmark
Thermal network	Cells, heat paths, overflow	Loss, body and mixing coefficients	Decision model
Flow solver	Velocity and buoyancy	Turbulence closure, jet data	Not used

2 Physical assumptions and scenario inputs

Each assumption below is paired with its reason and, where it matters, the scenario that tests it. Coefficients cannot be identified from the problem statement alone, so they come from the anchors in Section 2.1.

Table 3 separates assumed inputs from anchored coefficients.

Table 3. Baseline inputs

Quantity	Baseline / units	Status
Tub $L \times W \times H$	$1.50 \times 0.65 \times 0.23$ m	Assumed geometry
Displaced body volume / area	$0.060 \text{ m}^3 / 1.15 \text{ m}^2$	Assumed
Water density / heat capacity	$1000 \text{ kg/m}^3 / 4180 \text{ J/(kg K)}$	Rounded constants
Air / skin / inlet temperature	$22 / 34 / 50^\circ\text{C}$	Fixed reservoirs
Surface / wall / body coefficient	$25 / 6.5 / 25 \text{ W/(m}^2 \text{ K)}$	Correlation-based (next page)
Mixing diffusivity D	$0.001 \text{ m}^2/\text{s}$	Uncalibrated closure; range tested
Initial / lower / upper limit	$40 / 39 / 41^\circ\text{C}$	Preference scenario
Time / allowed spread	$1800 \text{ s} / 1.5^\circ\text{C}$	Preference scenario

A1. Constant density and heat capacity. Water properties change by under 1% between 39 and 41°C , and the constants are rounded values, not an evaluated IAPWS table [3]. *Reason:* the resulting capacity error is small next to the loss coefficients.

A2. Heat exchange proportional to temperature difference. Surface, shell and body exchange follow Newton's law with effective coefficients [2]. *Reason:* the temperature span is narrow, so linearization is acceptable. The surface coefficient aggregates convection, radiation and evaporation, so evaporation is not added twice; evaporative volume change is neglected. The evaporative part is linearized about the room temperature; an independent review estimated that this understates its sensitivity near 40°C by several percent (about 7% on the total), which is small next to the tested range.

A3. Skin held at a fixed temperature. *Reason:* thermoregulation is outside the problem; skin temperature is varied (32 and 36°C) instead of resolved.

A4. Inlet-to-overflow surface stream. The hot water travels along a prescribed surface path to the overflow, so short circuit is possible. The stream runs along the row of cells nearest the centre line (half a cell off it on meshes with an even number of rows). *Reason:* a different inlet geometry would need its own transport network and evidence.

A5. Mixing as one effective diffusivity D . Motion enters through D , with a separate surface-loss scenario. *Reason:* molecular diffusion alone would not represent circulation; D is anchored only by an order-of-magnitude scaling (Section 2.1) and is tested over a range; vertical mixing is tested separately in Section 9.

2.1 Where the coefficients come from

No temperature record exists to fit, so each coefficient is derived from a textbook relation or tied to a published measurement where one exists, and given a range; where nothing exists (mixing, the exposed fraction, the air and skin temperatures) the value is an assumption and is labelled as one. Surface loss sums natural convection above a hot horizontal surface ($Nu = 0.15 Ra^{1/3}$ for $10^7 < Ra < 10^{11}$, length A/P [5]; here $Ra = 1.9 \times 10^7$), linearized radiation (emissivity 0.96) and evaporation by the Lewis analogy. For open water at 40°C in 22°C air at 50% humidity the parts are 4.7, 6.1 and 24.6 W/(m² K) (0.66 kg/m² h evaporated), 35.4 in total. A bather covers part of the surface; with an assumed exposed fraction of 0.7 the central value is 24.8, and we use 25 within the range 17–37, which spans exposed fractions of 0.5 to 1.0. Evaporation correlations differ by tens of percent, so this is a scenario range, not a measured one.

The shell coefficient is a series resistance: a 5 mm shell (0.19 W/(m K)) with film coefficients of 300 inside and 8 W/(m² K) outside gives 6.5, and 4.7–8.4 for other thicknesses. These are typical engineering values, not measurements of a particular tub. The body coefficient is anchored by a measurement: Menzies et al. report a rectal-temperature rise of $0.9 \pm 0.3^\circ\text{C}$ after 30 minutes in 40°C water to the shoulders [6]. Taking the mean body mass of the study participants, 73 kg, $c = 3470 \text{ J/(kg K)}$ and a mean body rise of 1–2°C, the average heat uptake is 141–281 W, equal to 20–41 W/(m² K) with skin held at 34°C. We use 25, because a fixed skin node overstates the driving difference later in the bath. The mean body rise and the driving difference are assumptions, so this is an order-of-magnitude anchor, not a calibration. Holding the skin at a fixed temperature also ignores the heat capacity of the body, about 37% of the water's here ($73 \text{ kg} \times 3470 \text{ J/(kg K)}$) against 687 kJ/K): uptake is anchored only on its 30-minute average. Its decay is omitted in the baseline and tested with a finite effective reservoir in Section 11.1.

Mixing has only a scaling anchor. A mixing-length estimate $D \approx 0.1 u' \ell$, with a velocity scale u' and eddy size ℓ , gives about $1 \times 10^{-4} \text{ m}^2/\text{s}$ for buoyancy-driven flow alone ($u' = 0.02 \text{ m/s}$, $\ell = 0.05 \text{ m}$), 1×10^{-3} for gentle movement of the bather (0.1 m/s, 0.1 m) and 6×10^{-3} for vigorous movement (0.3 m/s, 0.2 m). We use 0.001 and test 0.0003–0.003. This is a scaling argument, not a measurement, so D stays a scenario.

The 39–41°C window is a preference; immersion studies use 40–42°C water [6], which is not a safety standard. The inlet cell is a point source: the same policy gives a maximum inlet-cell average of 40.55, 40.76 and 41.00°C on the three meshes of Section 11, because a smaller cell mixes less water with the hot inflow. A limit imposed on that cell would test the mesh, not the bath. The ceiling and the spread limit therefore apply outside a jet-mixing zone of 0.15 m radius around the inlet, on the assumption that a bather does not sit in the jet; the floor applies to every cell, which the energy bound of Section 4 needs. With this definition the constrained temperatures agree across meshes (Section 11).

Table 4 summarizes the coefficient anchors and scenario ranges.

Table 4. Coefficient anchors and the values used

Coefficient	Relation or anchor	Derived value	Used (range tested)
Surface, W/(m ² K)	Convection + radiation + evaporation [5]	35.4 open; 24.8 at 70%	25 (17–37)
Shell, W/(m ² K)	Series resistance, typical shell values	6.5 (4.7–8.4)	6.5 (4.5–8.5)
Body, W/(m ² K)	Uptake implied by core rise [6]	20–41 at fixed skin	25 (12–40)
Mixing D , m ² /s	Mixing-length scaling	1×10^{-4} to 6×10^{-3}	0.001 (0.0003–0.003)

3 A transparent well-mixed benchmark

Table 5 defines storage, transport and exchange symbols.

Table 5. Symbols

Symbol	Meaning
C	Heat capacity of the water, J/K
H_a, H_b	Air/shell and body conductance, W/K
T_a, T_b, T_{in}	Room, skin and inlet temperature, °C
T_{min}, T_{max}	Lower and upper limit, °C
q, J	Inlet flow, m ³ /s; added water, L
D	Effective mixing diffusivity, m ² /s

Let C be total water heat capacity, H_a the air/shell conductance and H_b the body conductance. A well-mixed overflow has the same temperature as the bath. Integrating the physical energy balance gives

$$C\dot{T} = H_a(T_a - T) + H_b(T_b - T) + \rho c_p q(T_{in} - T) \quad (3)$$

$$C = \rho c_p (LWH - V_b), \quad H_b = h_b A_b \quad (4)$$

$$H_a = h_s f L W + h_w \{L W + 2 H (L + W)\} \quad (5)$$

The foam multiplier f equals one without a layer. Setting $q = 0$, define $H = H_a + H_b$ and $T_e = (H_a T_a + H_b T_b)/H$. The cooling solution and first time to reach the lower limit are

$$T(t) = T_e + (T_0 - T_e)e^{-Ht/C} \quad (6)$$

$$t_c = \frac{C}{H} \log \frac{T_0 - T_e}{T_{\min} - T_e} \quad (7)$$

Baseline conductances are $H_a = 37.14$ W/K and $H_b = 28.75$ W/K, and $C = 686,565$ J/K. Thus $T_e = 27.24^\circ\text{C}$ and $t_c = 14.17$ min.

A heat-loss model must approach an environmental equilibrium rather than zero Celsius. This analytic boundary also supplies a direct check on signs and units. The benchmark describes mixing perfectly; the spatial model will test the cost of departing from that assumption.

4 What can be proved about water use?

Let $\ell(T) = H_a(T - T_a) + H_b(T - T_b)$ be the heat loss of the mixed bath, and let the inlet be hotter than every temperature considered. **Proposition 1.** Coasting to the lower limit and then holding it uses the least water, provided the holding rate is within the faucet bound. The proof has four steps.

Proof. **Step 1, an identity.** Dividing the balance by $T_{\text{in}} - T$ and integrating over the horizon gives

$$\int q \, dt = \frac{C}{\rho c_p} \log \frac{T_{\text{in}} - T_0}{T_{\text{in}} - T_f} + \int \frac{\ell(T)}{\rho c_p (T_{\text{in}} - T)} \, dt \quad (8)$$

Step 2, monotonicity. The integrand $\ell(T)/(T_{\text{in}} - T)$ has derivative $[H_a(T_{\text{in}} - T_a) + H_b(T_{\text{in}} - T_b)]/(T_{\text{in}} - T)^2 > 0$, so it increases with T , and the logarithmic term increases with the final temperature. Water use is therefore an increasing functional of the temperature path.

Step 3, the lowest admissible path. Because $q \geq 0$, comparison with the no-flow solution gives $T(t) \geq T_c(t)$, the cooling curve of Section 3, and feasibility requires $T(t) \geq T_{\min}$. The pointwise lowest admissible path is $\max(T_c(t), T_{\min})$: coast until T_c reaches $T_{\min}$, then hold.

Step 4, attainability. Holding $T = T_{\min}$ means $\dot{T} = 0$, which needs the rate

$$q_{\text{hold}} = \frac{\ell(T_{\min})}{\rho c_p (T_{\text{in}} - T_{\min})} \quad (9)$$

When this rate is within the bound, the path is feasible and, by Step 2, optimal.

□

For the baseline, the mixed-model optimum adds 16.01 L: wait 14.17 min, then supply 1.012 L/min. Optimality is established for the mixed model; spatial policies are assessed separately.

Proposition 2 (energy lower bound). In the spatial model, suppose every cell satisfies $T_i(t) \geq T_{\min}$. Summing the cell balances gives $\sum_i C_i T_i(t_f) - C T_0 = - \int L \, dt + \rho c_p \int q(T_{\text{in}} - T_{\text{out}}) \, dt$, with L

the total loss. Each loss term increases with its cell temperature, and H_a and H_b are the sums of the cell conductances, so $L(t) \geq \ell(T_{\min})$. The outlet cell satisfies $T_{\text{out}} \geq T_{\min}$, so $T_{\text{in}} - T_{\text{out}} \leq T_{\text{in}} - T_{\min}$, and the left side is at least $C(T_{\min} - T_0)$. Combining these three facts,

$$J \geq \max \left(0, \frac{1000[\ell(T_{\min})t_f - C(T_0 - T_{\min})]}{\rho c_p (T_{\text{in}} - T_{\min})} \right) \quad (10)$$

The resulting 15.41 L is a genuine conditional lower bound, valid for any control, but it is not tight: the initial water is hotter than the floor and loses more heat, and limited mixing adds further cost. Section 7.1 shows how much of the gap an optimized schedule recovers.

5 A conservative model in three dimensions

The rectangular water envelope is divided into $8 \times 4 \times 3$ cells. Each has a capacity C_i and exchanges heat only across common faces. The cells whose centres lie within 0.15 m of the inlet cell form the jet zone (2, 9 and 22 cells on the $8 \times 4 \times 3$, $12 \times 6 \times 4$ and $16 \times 8 \times 6$ meshes); Ω is the rest. Top cells lose heat to the room; bottom and side faces lose heat through the shell. The bather is represented by a smooth, three-dimensional displacement field and a distributed skin contact term.

$$C_i \dot{T}_i = \sum_j g_{ij}(T_j - T_i) + H_{a,i}(T_a - T_i) + H_{b,i}(T_b - T_i) + S_i \quad (11)$$

$$g_{ij} = \rho c_p D \frac{A_{ij}}{d_{ij}} \min(\phi_i, \phi_j), \quad C_i = \rho c_p V_i \quad (12)$$

Here ϕ is fluid fraction, A_{ij} a face area and d_{ij} the center separation. Symmetric g_{ij} guarantees that internal heat exchange cancels when the equations are summed. D is an effective mixing closure; molecular diffusion alone would not represent motion or buoyant circulation.

$$w_i \propto \exp \left[-\frac{1}{2} \sum_{k=1}^3 \left(\frac{x_{ik} - b_k}{s_k} \right)^2 \right], \quad \sum_i w_i = 1 \quad (13)$$

$$V_i = V_{\text{cell}} - V_b w_i, \quad H_{b,i} = h_b A_b w_i \quad (14)$$

Baseline $\mathbf{b} = (0.55L, W/2, 0.45H)$, with widths $\mathbf{s} = (0.40, 0.16, 0.12)$ m. Fluid fraction is $\phi_i = V_i/V_{\text{cell}}$. The center $\mathbf{b}$ and widths $\mathbf{s}$ set the spatial distribution of body effects. Body volume and contact area remain independent inputs: shape is varied through $\mathbf{s}$ at fixed volume and area. This is a homogenized immersed-body representation, not an anatomically resolved obstruction. A cell whose occupied fraction reaches 0.95 is rejected rather than assigned a negative capacity.

A fixed envelope volume is preserved by balancing inlet and overflow. In the baseline the exact displaced-water volume is 164.25 L. Shape scenarios preserve envelope volume when isolating aspect-ratio effects. Volume scenarios deliberately change water depth and hence both storage and side-wall area.

5.1 Spatial evidence: the mean is not the whole bath

Figure 1 locates cold regions across layers on one scale.

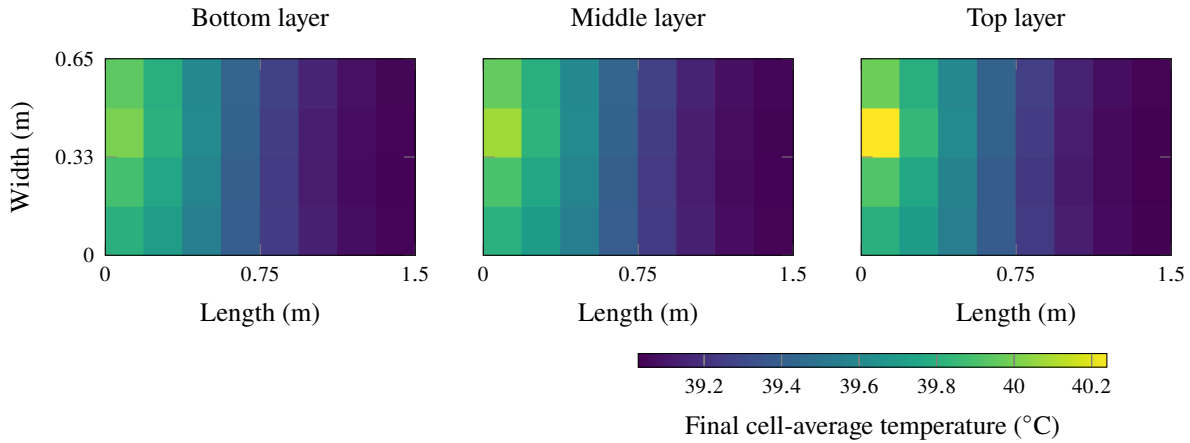


Figure 1. Final temperatures in the three horizontal cell layers. All layers use one color scale; the top layer contains the prescribed inlet-to-overflow stream. Geometry is in metres, and values are cell averages.

The heat map and Figure 2 use the same archived trajectory. Exchange couples the horizontal layers into one three-dimensional network. The cold-region locations explain information lost by a mean or range; the mesh table tests resolution. These are homogenized cell averages, not resolved anatomy, recirculation or a faucet jet.

6 Inlet transport, solver and strategy search

Inlet and overflow are on opposite ends of a surface stream. Every path edge carries the same q : the first cell receives hot water, interior cells receive upstream water and lose the same volume downstream, and the last cell discharges through the overflow.

$$S_1 = \rho c_p q (T_{in} - T_1), \quad S_i = \rho c_p q (T_{i-1} - T_i) \quad (15)$$

$$\sum_i C_i \dot{T}_i = - \sum_i H_{a,i} (T_i - T_a) - \sum_i H_{b,i} (T_i - T_b) + \rho c_p q (T_{in} - T_{out}) \quad (16)$$

This stream deliberately allows hot-water short circuit. Heat can leave at a locally warm outlet before warming a distant cold region. It is an explicit transport assumption, not an inferred flow field. Section 11.1 changes the internal path while preserving inlet and overflow; alternative inlet placement would still require another network and calibration.

For a fixed flow, the network is affine linear. Augmenting the state by a constant one permits matrix-exponential propagation, avoiding a large forward-Euler step restriction [4]. A delayed-start policy has two exact constant-control segments.

$$\dot{\mathbf{T}} = A(q)\mathbf{T} + \mathbf{b}(q), \quad \mathbf{z}(t + \tau) = e^{\tilde{A}(q)\tau} \mathbf{z}(t) \quad (17)$$

Starts are tested at 0, 120, ..., 1200 seconds. For each, rates from 0 to 3 L/min are bracketed on a 0.2 L/min grid; the first crossing of the lower-temperature requirement is refined by Brent's root method. The search targets $T_{\min} + 0.03^{\circ}\text{C}$ as numerical reserve, then checks the upper-temperature and spread limits at 5-second output intervals. Starts that already violate the floor are rejected.

This is a reproducible candidate search. Nonmonotone flow responses, untested delays, pulsed inputs, inlet relocation and feedback are not excluded by the calculation. A failed search means no candidate was accepted, not that every possible action is infeasible. The independent checks further examine the chosen trajectory between samples.

7 The spatial result and the mixing penalty

Figure 2 contrasts the mean, coldest cell and spatial range.

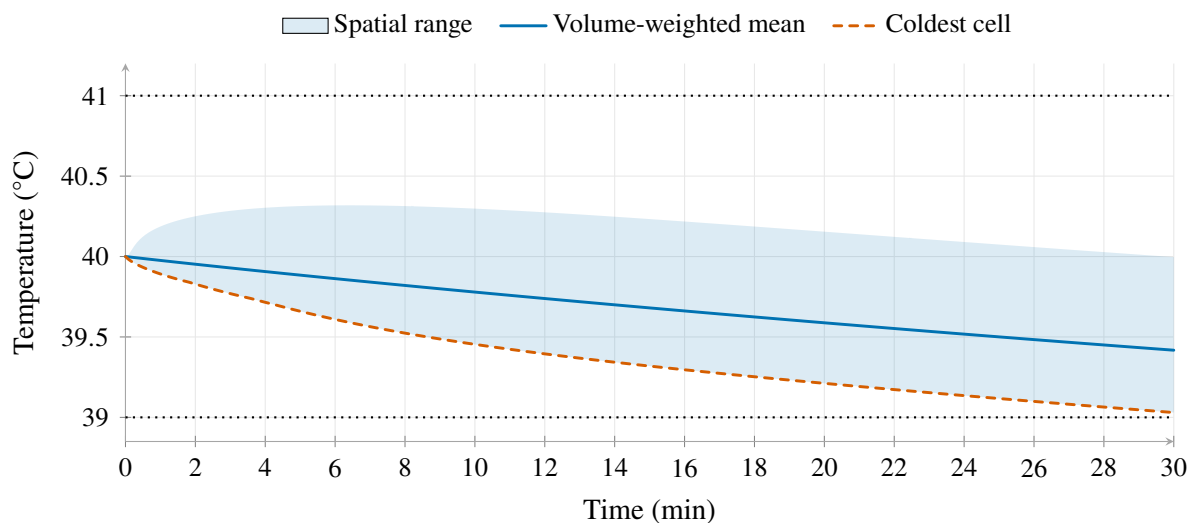


Figure 2. Matrix-exponential cell temperatures under the selected policy. The shaded range covers the coldest cell and the hottest cell outside the inlet jet zone; the mean alone would hide cold and hot locations.

Table 6 reports the constant-rate trajectory and water use.

Table 6. Baseline result under the best constant rate

Result	Baseline
Rate / start	0.805 L/min / 0.0 min
Added water	24.14 L
Minimum / maximum	39.030 / 40.318°C
Maximum simultaneous spread	0.967°C
Final volume-weighted mean	39.418°C

Within the constant-rate family, starting the trickle immediately ranks ahead of delayed starts here, even though waiting is optimal in the ideal mixed model. The distant water needs time to re-

ceive heat; larger late rates create a warmer inlet region before they solve the cold-region constraint. This is a concrete consequence of resolving space.

Holding a perfectly mixed bath at 40°C throughout would use 36.22 L. This is a stricter reference service, not a like-for-like optimum. Our accepted 1°C cooling allowance uses less water partly because it provides a different service. The comparable mixed model with the same lower limit uses 16.01 L. The gap also reflects our restricted control family and 0.03°C numerical reserve; it cannot be attributed purely to imperfect mixing.

7.1 A time-varying schedule uses less water

Constant-rate policies are a narrow family. We therefore optimize piecewise-constant inlet flow in K equal segments ($K = 3, 6, 12$) by sequential quadratic programming, minimizing added water under the same floor and outside-jet limits (with the 0.03°C floor reserve), sampled every 5 s, from four starting schedules. The problem is non-convex: each result is the best local optimum found, with no global claim.

Table 7 separates candidates from proved bounds.

Table 7. Constant rate, scheduled flow and the bounds

Policy	Water (L)	Change	Min / max (°C)	Spread (°C)
Best constant rate	24.14	—	39.03 / 40.32	0.97
3 segments, coarse candidate	19.75	-18.2%	39.03 / 40.74	1.50
6 segments, coarse candidate	19.50	-19.2%	39.03 / 40.73	1.50
12 segments, coarse candidate	19.35	-19.9%	39.03 / 40.80	1.50
6 segments, accepted	21.48	-11.0%	39.13 / 40.79	1.42
Perfect-mixing optimum	16.01			
Energy lower bound	15.41			

```

Input: network, limits, K, starting schedules
for each starting schedule x0:
    minimize sum(x)*segment_time over 0 <= x <= 3 L/min
    subject to min_i T_i(t) >= Tmin+reserve,
               max_{i in Omega} T_i(t) <= Tmax,
               spread over Omega <= span (every 5 s)
retain coarse candidates and buffered alternatives;
replay on three meshes with independent integration;
accept only if continuous bounds meet all limits

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Figure 3 shows flow timing and temperature constraints.

The accepted 6-segment policy is off for the first 5.0 minutes and last 10.0 minutes, with a peak flow of 1.67 L/min. For consecutive 5-minute intervals the rates, rounded to nine decimals, are 0, 1.668107192, 1.343440056, 1.283871597, 0, 0 L/min; the interval duration times their sum is

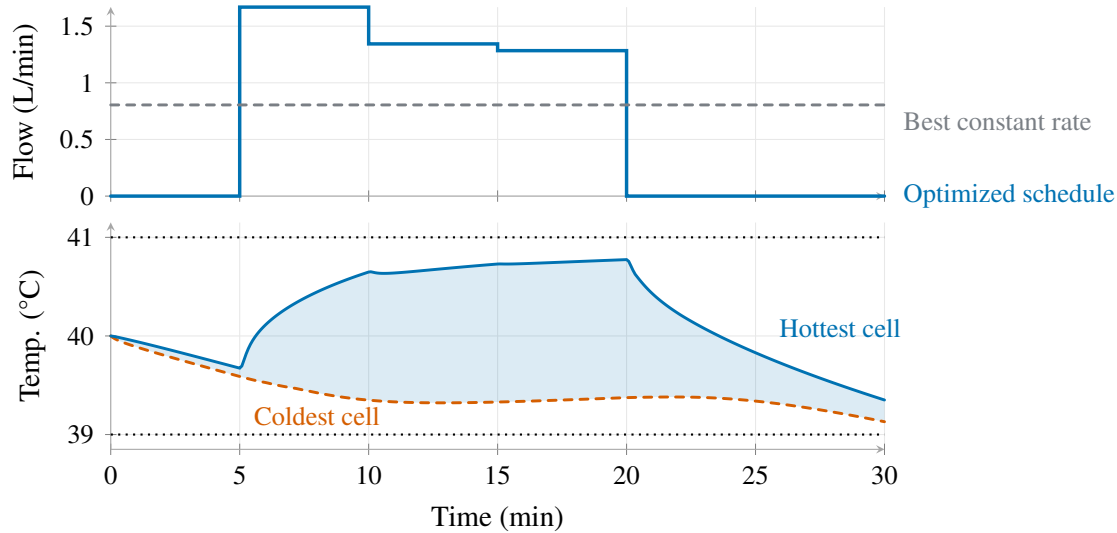


Figure 3. Selected buffered 6-segment schedule against the best constant rate (top) and the range of cell temperatures outside the inlet jet zone (bottom); dotted lines mark the limits. The baseline concentrates inflow in the middle 15 minutes; both no-flow periods enter the independent checks.

21.477094 L. It adds 21.48 L, 11.0% below the constant rate. Its 0.1°C design margin sacrifices some water savings to protect the stated constraints. The remaining 6.06 L gap to the energy bound is not an optimality certificate.

The unbuffered 12-segment candidate uses 19.35 L but reaches a spread of 1.531°C on the finest mesh, above the 1.5°C limit, so it is rejected. Among the tested buffered alternatives, the accepted 6-segment policy gives spreads of 1.400, 1.395 and 1.416°C on the $8 \times 4 \times 3$, $12 \times 6 \times 4$ and $16 \times 8 \times 6$ meshes. Independently integrated continuous-time bounds also pass on each mesh. This establishes numerical feasibility for these tested networks, not mesh-independent physics or a global minimum.

7.2 What does staying close to the start temperature cost?

The task asks for a bath close to its initial temperature without wasting much water, which is a trade-off rather than a single optimum. Figure 4 repeats the search while the allowed fall below 40°C changes from 0.25 to 3°C ; the 1°C case is the baseline. With a fall of 2.5°C or more the stored heat suffices and no water is added. At 2°C the best constant rate adds 1.9 L, at 1.5°C 12.9 L, at 1°C 24.1 L and at 0.75°C 29.8 L. Between 2°C and 0.75°C each further 0.5°C of tolerance is therefore worth roughly 11 L, almost linearly. At 0.5°C the best constant-rate policy was accepted, while the perfectly mixed bath would still need 30.7 L.

Figure 4 compares searched spatial policies with analytical benchmarks.

The spatial-minus-mixed gap increases from 1.5 L at a 2°C fall to 9.2 L at 0.75°C . Scheduling lowers the baseline by 11%; schedules at other tolerances were not optimized.

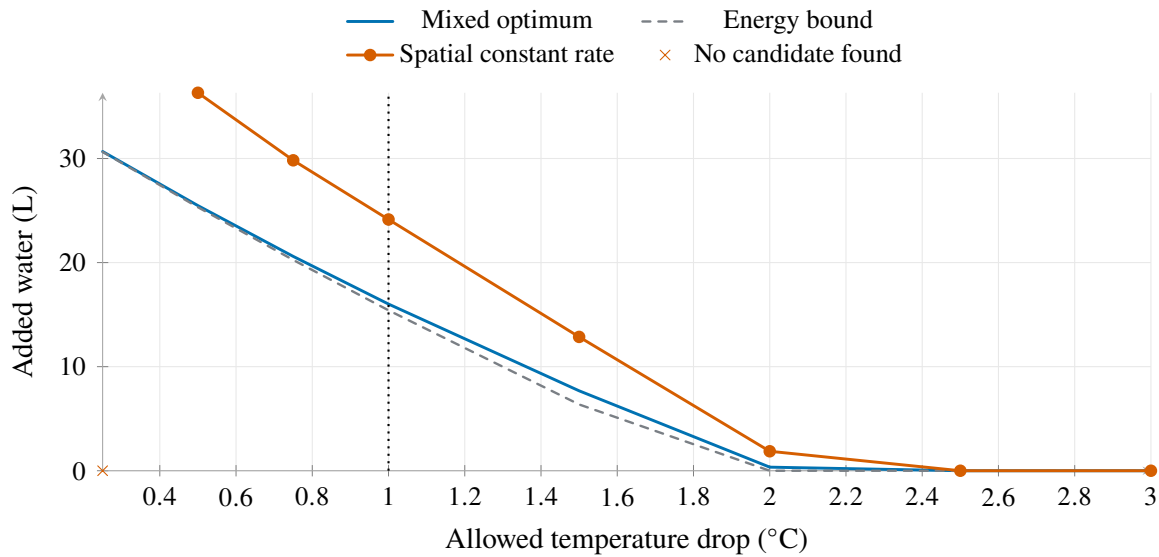


Figure 4. Water use versus temperature tolerance. Crosses denote unsuccessful searches, not zero water demand; the dotted line marks the baseline. Allowing a larger fall sharply reduces demand in the tested constant-rate family.

7.3 Can a user follow the schedule?

We distinguish fixed tap bias from random segment errors. For the unbuffered 12-segment candidate, multiplying every rate by 0.9 or 1.1 crosses a limit by 0.076 and 0.126°C. Independently drawing each multiplier from a normal distribution of mean 1 and standard deviation 0.1, clipped below zero, gives median and 5th-percentile slacks of -0.070 and -0.217°C (200 draws, seed 11). Standard deviation 0.1 does not mean an error bounded within $\pm 10\%$. Negative slack is a violation, however small.

The error is not symmetric. The most sensitive segment is segment 4 (2.81 L/min), where raising the flow by 20% breaks the limits by 0.245°C and lowering it by 20% by 0.047°C: extra hot water overheats the inlet region and widens the spread. Over-delivery in a high-flow segment is the error to guard against. 1 of the 5 active segments tolerates a 20% error in either direction.

The price of a margin is measured on a 6-segment version. Requiring all three limits to hold with a buffer b gives 19.50 L at $b = 0$, 21.48 L (+10%) at 0.1°C, 23.81 L (+22%) at 0.2°C and 26.83 L (+38%) at 0.3°C. With the 0.1°C buffer, 74% of 200 draws with independent Gaussian multiplier standard deviation 0.1 stay within the limits. The 0.1°C policy is selected here for its independently checked numerical margin. Its 74% acceptance under the stated random error model does not make it a reliable manual prescription; a real bath needs temperature feedback and separate calibration.

7.4 Measurement ambiguity changes the decision

Mixing changes timing: the weaker-mixing candidate starts immediately. Two probes miss violations in two of nine cases. We now propagate observation ambiguity into the supply decision.

A separate synthetic trial applies (0, 0, 1.2, 0, 0, 0) L/min in five-minute stages. Known geometry starts uniformly at 40°C; cells 89 and 6 are read every 30 s. Assumed reading errors and constant probe biases are each bounded by $\pm 0.02^\circ\text{C}$; these are not instrument specifications.

For parameters θ and probe j , let $d_j(t; \theta) = T_j(t; \theta) - y_j(t)$. At observed times $\mathcal{T} = \{0, 30, \dots, 1800\}$ s, a model is compatible when a constant correction a_j exists such that

$$\min_{|a_j| \leq 0.02} \max_{t \in \mathcal{T}} |d_j(t; \theta) - a_j| \leq 0.02 \quad (j = 1, 2). \quad (18)$$

The extreme-discrepancy midpoint clipped to the bias bound is the minimax correction, with physical bias $b_j = -a_j$. A trace from -0.04 to +0.04°C fails: bias cannot vary between readings.

Our grid has 7 mixing values (0.00085–0.00115 m²/s), 9 surface and 9 body coefficients (20–30 and 15–35 W/(m² K)), and 5 delivery multipliers (0.95–1.05). Of 2835 models, 178 fit. This finite decision ambiguity set is neither a confidence region nor a probability distribution.

Table 8 compares archived policies with a replacement optimized on ten extreme/failing models, using a 2 L/min bound, 0.1°C target and 0.03°C extra floor reserve. All 178 pass three-grid envelopes; the tightest spread bound is 1.496828°C. The selected target is not an all-model margin or global optimum.

Table 8. Policies on the same measurement-compatible set

Policy	Command (L)	Delivered (L)	Sampled pass
Constant	24.14	22.94–25.35	130/178
Nominal six-stage	21.48	20.40–22.55	174/178
Ambiguity-set six-stage	23.48	22.31–24.66	178/178

The table uses the same 178 models and five-second sampling; archived policies were not re-optimized. Only the replacement has 534 three-grid envelopes. Its five-minute rates are 0.892820, 1.473786, 1.179268, 1.150408, 0.000000, 0.000000 L/min; envelope extrema are 39.1297/40.9238°C.

The replacement costs 9.33% more than the nominal policy. A separate zero-inlet trial with the same prior, probes and errors retains 510 models, including unidentified delivery bias; its candidate passes 1530 three-grid envelopes. The two observation sets intersect in 167 models; they are not nested or probabilistic.

Table 9 separates control and trial water. Both candidates restart at uniform 40°C with the same control family and design targets.

Passive rates are 1.419252, 1.083923, 1.055204, 1.044908, 0.274962, 0.000000 L/min. Initial filling and reset are not free; these conditional candidates do not establish expected information value or global optimality.

Let n unchanged baths share one trial; let $\alpha \in [0.95, 1.05]$ be the common delivery multiplier and R_0, R_1 the actual reset litres for passive and pulse designs. The difference between their total

Table 9. Observation-conditioned trial and control costs

Observation / trial	Models	Control (L)	Trial (L)
Nominal / passive	510	24.39	0.00
Nominal / pulse	178	23.48	6.00
Strong mixing / passive	60	16.51	0.00
Strong mixing / pulse	14	15.37	6.00
Weak mixing / passive	70	Not accepted	0.00
Weak mixing / pulse	11	35.05*	6.00

actual-water costs is

$$C_0 - C_1 = \alpha(n\Delta - 6) + R_0 - R_1, \quad \Delta = 0.909836 \text{ L.} \quad (19)$$

At nominal delivery, passive costs less for one bath if $R_0 - R_1 < 5.0902 \text{ L}$. Equal reset costs give a crossover at 6.59 baths: passive is cheaper through six, pulse from seven. Reuse requires unchanged geometry, heat transfer, probes and delivery bias; otherwise remeasure. This is not online feedback or a real-bath guarantee.

Two additional observations use $(D, \text{ surface/body coefficients})=(0.00115, 20/15)$ and $(0.00085, 30/35)$, in m^2/s and $\text{W}/(\text{m}^2 \text{ K})$, with nominal delivery. Separate trials retain the same prior/errors. Strong mixing gives a 1.1406 L control-cost gap, so equal-reset pulse reuse pays from six. The three accepted candidates pass 255 distinct three-grid envelopes.

*The weak-mixing pulse passes physical limits but fails the optimizer and extra design margin. No passive candidate was accepted within two starts and three constraint-generation rounds; neither failure proves infeasibility. These are observation-conditioned outcomes, not *ex ante* experiment values or sequential inference.

8 Separate geometry, size and body effects

Table 10 isolates geometry, storage and body changes.

At equal volume, the wide, shallow tub has a larger exposed top area and a changed diffusion length; it has no accepted candidate. The deep, narrow tub needs 17.48 L against 24.14 L, because a deep, narrow envelope reduces top-area heat loss, although a real deep bath may stratify, which Section 9 tests with a separate vertical-mixing scenario. Geometry is therefore more than an interchangeable cooling coefficient.

Changing depth at fixed length and width changes thermal storage and shell area. Greater water volume slows cooling, which can lower added water over a short fixed horizon: the large bath needs 20.39 L, while the small bath needs 27.05 L. The 0.23 m baseline depth is shallow, and its surface-to-volume ratio is about 1.7 times that of a bath filled to the shoulders; a 0.35 m depth needs 13.05 L, so the baseline overstates the surface-loss share of a deeper bath. That does not mean a larger bath conserves total water: filling it initially uses more. J counts replenishment only, and this distinction

Table 10. Geometry, size and body scenarios

Change from baseline	Water (L)	Interpretation
Shallow Wide	No candidate	Same envelope volume
Deep Narrow	17.48	Same envelope volume
Small Bath	27.05	Lower water depth
Large Bath	20.39	Higher water depth
Deep Bath	13.05	Depth 0.35 m, like a bath filled to the shoulders
Larger Body	28.11	More displacement/contact
Long Body	23.85	Shape only; fixed volume/area
Warmer Skin	21.08	Skin at 36°C
Cooler Skin	27.18	Skin at 32°C

prevents a misleading conservation claim.

The larger-body scenario simultaneously displaces more water and increases assumed contact area; it is a combined size scenario and needs 28.11 L. The long-body scenario isolates the spatial distribution by keeping total body volume and contact area fixed. Warmer skin reduces the temperature difference driving body heat loss. These changes quantify conditional dependence; they do not establish measured human heat transfer.

To transfer the model to a real tub, measure water volume after entry, submerged contact geometry, temperatures at several depths and distances, and no-inlet cooling. Shape and temperature should not be inferred from a single mean cooling curve.

9 Motion and a bubble layer can change the policy

Table 11 separates transport benefits from boundary losses.

Table 11. Mixing, surface, comfort and supply scenarios

Scenario	Added water (L)	Search outcome
Weak Mixing	—	None accepted
Strong Mixing	19.40	Accepted
Moving With Added Surface Loss	23.04	Accepted
Stratified	24.42	Accepted
Convective Surface Layer	24.10	Accepted
Foam	10.37	Accepted
Tight Comfort	36.32	Accepted
Loose Comfort	1.87	Accepted
Low Loss	8.60	Accepted
High Loss	—	None accepted
Cool Supply	44.80	Accepted

Increasing D reduces the gradient created by the localized inlet. The strong-mixing scenario uses 19.40 L against 24.14 L, but motion may also increase heat loss. When D is increased together with a 20% increase in the surface coefficient (30 instead of 25 W/(m² K)), the need rises to 23.04 L and most of the benefit disappears. The comparison deliberately separates transport improvement from its possible boundary cost.

Stratification is tested separately. A hot inlet layer is lighter than the water below, and the stable density gradient suppresses vertical mixing. With a cell height of 0.077 m, a vertical temperature difference of 1 K, a velocity scale of 0.02 m/s and an expansion coefficient of about 3.8×10^{-4} per K, the gradient Richardson number is $Ri = g\beta \Delta T \ell / u^2 \approx 0.7$. The Munk–Anderson stability function for scalars, $(1 + 3.33 Ri)^{-3/2}$ [8], then reduces vertical diffusivity to about 0.15 of its neutral value. We test a vertical-to-horizontal ratio of 0.2. The best constant rate becomes 24.42 L against 24.14 L, but the maximum spread rises from 0.97 to 1.07°C against the 1.50°C limit. Stratification therefore costs little water here and uses 19% of the remaining uniformity margin; the ratio is an order-of-magnitude scenario and the Richardson estimate uses assumed velocity and temperature scales. The stable layer is an upper-bound picture: evaporative cooling at the surface can instead drive convection and raise vertical mixing, so a vertical-to-horizontal ratio of 3 is also tested, and it needs 24.10 L.

The foam scenario multiplies only the effective surface coefficient by 0.4. With all other inputs fixed, the best constant rate falls to 10.37 L. The assumed 60% reduction is a scenario, not an experimentally established property of bubble-bath additive. If the layer breaks up or motion raises evaporation, this result must be recomputed.

No accepted candidate exists for: Weak Mixing, Shallow Wide, High Loss. They are reported as such, not hidden in a favorable average. In the weak-mixing, high-loss and wide-shallow cases, flows large enough to keep every cell above the floor already push the spread above its 1.5°C limit (checked at constant flows from 0.8 to 2.5 L/min), but these sampled candidates cannot establish that every constant flow fails. No constant-rate policy was accepted by the documented search in those scenarios. Their outcomes identify where the restricted strategy must change. Potential responses include improved circulation, a different inlet path, a shorter bath, or a different comfort tolerance. The paper does not certify which response is safe or optimal without corresponding physical evidence.

10 Validation: independent evidence and remaining error

All 17 recorded checks pass. Independent heat-flow arithmetic sums environmental losses, body exchange and local-temperature overflow; the largest instantaneous residual is 2.46×10^{-10} W. An adaptive RK45 integration with independently assembled right-hand side agrees with the archived matrix-exponential trajectory within 3.12×10^{-07} °C. This tests numerical implementation, not a second physical model.

A constructed uniform-loss problem reproduces its analytic exponential cooling solution. Internal exchange cancels globally, and the stream has exactly one overflow sink. Positive off-diagonal

dynamics and nonpositive row sums support physical bounds and a contractive derivative estimate for each constant-flow segment.

Half-second replay is supplemented by a derivative bound: if $|dT_i/dt| \leq L$, a point lies at most 0.25 seconds from a sample, so temperatures differ by at most $L/4$ and spread by at most $L/2$. With an integration allowance and no relaxation of the physical limits, the resulting lower envelope is 39.0144°C, upper 40.3340°C, and spread upper bound 0.9980°C. This is a conditional floating-point envelope, not interval-arithmetic certification.

The accepted schedule uses an independently assembled right-hand side and adaptive RK45, restarted at every flow switch. Nonnegative off-diagonal transport and nonpositive row sums give an infinity-norm contraction bound for the derivative within each segment. One-second samples alone leave some intervals unresolved; adding RK45 step endpoints and bisecting unresolved intervals once on the finest mesh gives bounds of 39.1297°C for the minimum, 40.8304°C for the maximum and 1.4964°C for the spread. These pass the physical limits without threshold relaxation. A 2×10^{-6} °C integration allowance is included; this floating-point check is conditional on the network, not interval arithmetic or experimental validation. The 17 baseline checks are separate.

11 Resolution, uncertainty and transfer to a real bath

Table 12 compares constrained temperatures and the excluded inlet peak.

Table 12. Mesh replay: constrained temperatures

Mesh (cells)	Max outside jet zone	Spread outside zone	Inlet-cell max
$8 \times 4 \times 3$ (96)	40.318°C	0.967°C	40.553°C
$12 \times 6 \times 4$ (288)	40.315°C	0.960°C	40.757°C
$16 \times 8 \times 6$ (768)	40.330°C	0.973°C	40.995°C

The same constant-rate policy is replayed on three meshes. The ceiling and spread quantities outside the jet zone agree within 0.015°C and 0.013°C, and the tested constant-rate trajectory stays within the comfort limits on these meshes. The inlet-cell maximum rises from 40.55 to 41.00°C with refinement; that is why the cell is excluded rather than constrained. The selected water amounts on the $8 \times 4 \times 3$ and $12 \times 6 \times 4$ meshes are 24.142 and 24.117 L (0.11% apart). Three meshes are a diagnostic, not an asymptotic convergence-order study.

Parameter uncertainty concerns heat loss, body exchange and mixing. Structural uncertainty concerns the prescribed surface stream, fixed skin reservoir, homogenized body and constant effective D . These are different errors. The scenario table explores parameter dependence and one coupled motion effect; it is not a probability distribution or confidence interval.

Range analysis. Section 2 gives ranges for six inputs. 64 scrambled Sobol points cover them (D log-uniform), and the constant-rate search runs at each. 41 points have an accepted candidate. Among those, the required water is 12.5, 17.0, 22.0, 26.9 and 30.9 L at the 5th, 25th, 50th, 75th

and 95th percentiles. The remaining 23 points have none, mostly because of weak mixing (below). The baseline of 24.1 L sits in the upper half of this distribution.

Spearman rank correlations with water, over accepted draws: surface coefficient +0.63, room temperature -0.47, body coefficient +0.47, skin temperature -0.27, shell coefficient +0.15, mixing diffusivity -0.15.

Acceptance depends most strongly on mixing in this finite search: median D is $1.4 \times 10^{-03} / 4.5 \times 10^{-04}$ m²/s for accepted/rejected draws (Mann–Whitney $p < 1 \times 10^{-4}$); the other five inputs show no detected association (smallest $p=0.47$). Weak mixing can leave distant water cold while larger inflow violates uniformity. The chosen ranges describe scenarios, not real-bath probabilities.

An inlet-free trial cheaply characterizes cooling but cannot identify a term it never excites. At uniform initial T_0 , internal exchange vanishes, so $\dot{T}_i(0) = [H_{a,i}(T_a - T_0) + H_{b,i}(T_b - T_0)]/C_i$, independent of D . Later spatial differences can reveal mixing. Delivery bias and inlet path remain absent throughout the passive trial. A warming contact reservoir shares the initial slope but changes later curvature; loss parameters can compensate for that difference. An inlet trial must therefore check transport and overflow energy, not merely refine a cooling fit. Reserve a separate experiment to check predictions; match motion/foam trials for changes in loss as well as mixing.

The finite-volume stencil is also checked against an insulated three-dimensional cosine diffusion mode: four successively refined grids approach second order (finest observed order 1.98). A zero-diffusion inlet path matches the independent stirred-cell cascade step response within 4.3×10^{-14} °C. These verify implementation in empty-domain limits, not the mixing closure or real baths [9].

11.1 Carry structural ambiguity into the decision

The alternatives replace fixed skin by $C_b \dot{B} = \sum_i H_{b,i}(T_i - B)$ (253.31 kJ/K, initially 34°C), or route the inlet stream down, across the bottom and back to the same overflow. Contact storage is not physiology; the fixed-skin energy bound does not transfer.

Eight alternative-mechanism observation sets fit the original structure; six existing policies pass specified sampled replays, with min/max/spread/final-body summary differences below 1.85×10^{-08} °C. We therefore screen surface/fixed, deep/fixed and surface/finite hypotheses together: passive/pulse banks retain 1465/467 pairs. Each trial restarts uniformly; observations are not sequential.

Base-grid success hid two finest-grid ceiling failures (41.0010/41.0007°C). A 1% passive-flow reduction repairs those sampled cases, without common reserves or full fine-grid coverage. New six-stage candidates instead target 39.13°C floor, 40.9°C outside-zone ceiling and 1.4°C spread across all three grids. These extra reserves leave the physical 39/41/1.5°C limits unchanged.

Table 13 compares structure-specific banks at the same reserves.

The local search uses exact affine sensitivities on active cells. Two base-grid members fail the common floor despite passing physical limits. Adding those constraints gives the candidates in Table 13. For consecutive five-minute stages their rates are 1.303664, 1.053753, 1.011697, 0.988372, 0.860138, 0.000000 L/min (passive), and 1.126406, 1.318679, 1.149624, 1.130662, 0.000000,

Table 13. Common-reserve structural decisions

Structure / trial	Pairs	Control (L)	Floor / ceiling (°C)	Spread (°C)
Surface/fixed / passive	510	26.09	39.132 / 40.857	1.374
Surface/fixed / pulse	178	23.63	39.132 / 40.900	1.400
Deep/fixed / passive	510	26.09	39.132 / 40.900	1.400
Deep/fixed / pulse	149	23.63	39.133 / 40.900	1.400
Surface/storage / passive	445	26.09	39.141 / 40.848	1.374
Surface/storage / pulse	140	23.63	39.145 / 40.895	1.400

0.000000 L/min (pulse). Unrounded settings are retained for computation.

All 5796 model-grid objects pass conditional continuous-time envelopes. The dynamic temperatures $U = (T_1, \dots, T_N, B)$ exclude the constant coordinate. Nonnegative off-diagonals and nonpositive row sums give contraction within each flow stage. Chord error is bounded by $\|U''(t_0)\|_\infty h^2/8$; the first derivative gives another enclosure. Unresolved one-second intervals are bisected to at most 0.0625 s. A $2 \times 10^{-6}^\circ\text{C}$ allowance (twice for spread) is assumed, not interval-proved. Table 13 rounds outward. Independent RK45/energy checks cover 54 extremal identities, not the entire bank.

Section 7.4's cost identity now has $\Delta = 2.461267$ L, giving $n > 2.43777$: pulse beats the fixed passive candidate from use 3, if one 6 L trial is reused, delivery bias stays positive and fixed, and total reset costs match. The seven-use result was conditional on the original structure. This expanded-bank comparison establishes neither global optimality nor expected experiment value.

11.2 Use observations to change the next action

Starting from 1,465 passive-compatible pairs on the 96-cell grid, read probes 89 and 6 every minute. Given prediction $\hat{T}_{mj,k}$ and reading $z_{j,k}$, retain each model's constant-bias intervals:

$$I_{mj,k} = [-0.02, 0.02] \cap \bigcap_{\ell=0}^k [z_{j,\ell} - \hat{T}_{mj,\ell} - 0.020002, z_{j,\ell} - \hat{T}_{mj,\ell} + 0.020002]. \quad (20)$$

Exclude a model if either interval is empty. Choose the least of 0, 0.8, 0.9 and 1 times the current backup rate whose next minute and remaining backup satisfy all survivors' common targets. Chord bounds include the dynamic body state, refine unresolved five-second intervals to 0.0390625 s, and assume the same $2 \times 10^{-6}^\circ\text{C}$ allowance.

Conditional feasibility and water cap. Each accepted minute is checked with the original remaining backup. Subsequent observations only shrink the set, preserving that feasible tail. Induction applies if the truth stays in the static bank and propagators are valid. For L/min commands $0 \leq q_k \leq q_k^{\text{backup}}$, a completed service has $J_{\text{cmd}} = \sum_k q_k \Delta t \leq \sum_k q_k^{\text{backup}} \Delta t \leq 26.09$ L ($\Delta t = 1$ minute). A shared $\alpha > 0$ gives $J_{\text{act}} = \alpha J_{\text{cmd}} \leq \alpha \times 26.09$ L. This proves neither minimum water nor fault or timeout recovery.

Fixed probe biases (+0.013, -0.011)°C give 23.59 L with zero reading noise and 23.16 L with one bounded-noise sequence (seed 20261010); 114/1 models remain. Both start from 1,465 hypotheses and the 26.09 L backup. Figure 5 shows action changes before unique identification. These are two nominal surface/fixed examples, not population performance.

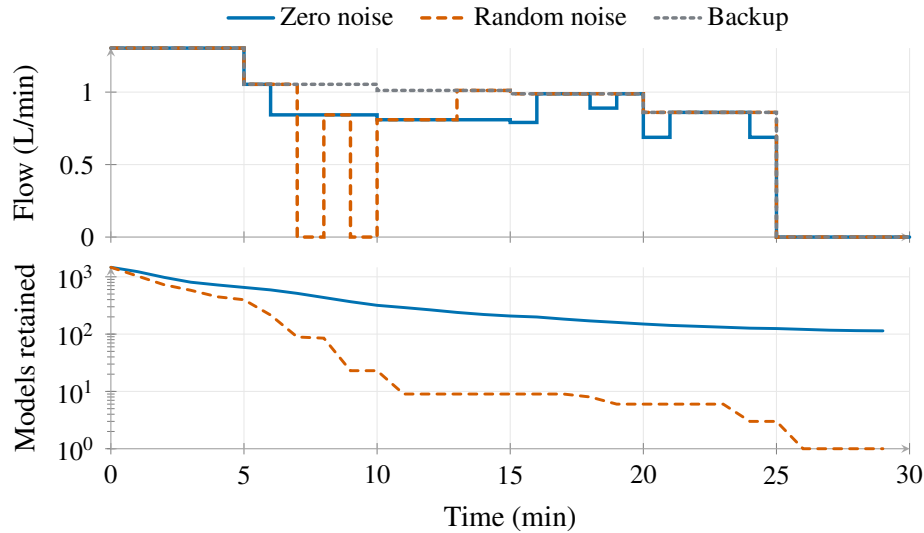


Figure 5. Observations change replenishment. Both nominal cases start from the same bank and backup; a logarithmic count axis distinguishes slow and rapid elimination. The constant bias and noise bounds are shared.

A separate matched 12-model ablation uses 26.00 L without observations versus 22.86/22.86/22.08 L with them in three structures. Six feedback trajectories pass 18 independent half-second three-grid RHS/energy replays; realized actions do not certify fine-grid feedback throughout the bank.

Minute-10 shifts to 45°C supply or 40 W/(m² K) surface loss empty the set at minute 11; terminal bias intervals independently exclude all 23 preceding survivors. Neither partial service counts as savings. An empty set invalidates reuse, without identifying the cause or certifying recovery.

11.3 Check transfer beyond the parameter grid

Six predeclared bank-external truths give two complete services (22.59/22.80 L commands). Changed storage, dimensions, supply or loss empties the bank after 28/5/11/9 minutes, before a checked temperature violation. Partial services are not savings; refusal supplies no recovery policy.

For the two completed surface/deep paths, fix the realized commands, baseline geometry, uniform 40°C water, 34°C fixed contact, 50°C inlet and 22°C air. Around $(D, h_s, h_b, \alpha) = (0.001025, 24.375, 26.25, 1.0125)$, the respective halfwidths are 6.25×10^{-06} m²/s, 0.15625 and 0.3125 W/(m² K), and 0.003125. Parameters are static; these are mathematical neighborhoods, not measured tolerances.

Fixed-command perturbation bound. Write $U' = M_\theta U + c_\theta$, affine in the four parameters. For halfwidths w_j , set $K_j = w_j \partial_j M$, $f_j = w_j \partial_j c$, and $\theta = \theta_0 + w \odot z$ with $|z_j| \leq 1$. Nominal variations satisfy $S'_j = M_0 S_j + K_j U_0 + f_j$, $S_j(0) = 0$.

Proof. The exact remainder $r = U_\theta - U_0 - \sum_j z_j S_j$ satisfies

$$r' = M_\theta r + \sum_{j,k} z_j z_k K_j S_k, \quad r(0) = 0. \quad (21)$$

Positive parameters give nonnegative off-diagonals and nonpositive row sums at every vertex, hence throughout the affine box. Contraction [10] and variation of constants give

$$\|r(t)\|_\infty \leq R(t), \quad R(t) = \sum_{j,k} \int_0^t \|K_j S_k(u)\|_\infty du. \quad (22)$$

The cellwise radius is $\sum_j |S_j| + R$: subtract it for the floor, add it for the ceiling, and twice for spread. States are continuous at flow switches; R accumulates across them.

□

Half-second derivative/chord bounds enclose the intervals; assumed errors are $2 \times 10^{-6}^\circ\text{C}$, $2 \times 10^{-6}^\circ\text{C/s}$ and $2 \times 10^{-6}^\circ\text{C/s}^2$ for states and first/second derivatives. The tested halfwidth scales were 4, 2, 1 and 0.4 times those above; the displayed box is the largest qualifying both paths on all three meshes: floor $\geq 39.0759^\circ\text{C}$, ceiling $\leq 40.7769^\circ\text{C}$, spread $\leq 1.1746^\circ\text{C}$. Common extra reserves do not all pass. Independent RHS/energy checks cover all 102 corners/centers; finite points challenge implementation, not prove coverage. A prior max-norm defect bound certified none. This exact-remainder result assumes numerical accuracy and certifies fixed actions, not all feedback branches or real baths.

12 Conclusions and a policy with clear scope

Choose tolerance, then check transport. Figure 4 prices warmth. Stored heat can defer inflow; weak mixing may require it immediately. Geometry and body contact change demand; motion redistributes heat but can raise loss, while foam reduces loss.

The baseline uses 24.14 L; the 6-stage policy uses 21.48 L with a 0.1°C design margin and three-grid checks. Common-reserve passive/pulse candidates use 26.09/23.63 L; a reusable 6 L pulse pays from use 3 with equal resets and unchanged conditions. Those fixed-candidate costs do not establish optimal feedback.

Match each guarantee to its scope. Static-bank feedback caps commands at 26.09 L and yields 23.59/23.16 L in the nominal noise cases. Section 11.3 extends two realized schedules to narrow static parameter boxes; it does not extend every feedback branch. Four changed-physics refusals invalidate reuse, without certifying recovery. Cell averages do not establish hot-jet safety. These conclusions require the stated initial state, transport and error conditions.

13 A warmer bath, with less replacement water

A guide for the person in the bathtub

Choose your tolerance. Allowing some cooling uses stored heat before replacement water. Our example needs about 24 L with a 1°C allowance, but 1.9 L with 2°C. These are model scenarios; your comfort and health needs determine the acceptable range.

Redistribute warmth. Water near the faucet may warm while distant water stays cool. Gentle movement can help; stronger motion can also increase surface heat loss. More inflow is not automatically better, because it may widen temperature differences or send useful heat straight to the overflow.

Use timing only under its stated conditions. The baseline schedule waits 5 minutes, supplies water for 15 minutes and stops for 10 minutes. It uses 21.48 L against 24.14 L for a constant trickle. Weaker mixing can require inflow immediately. Even with a margin, only 74% of the specified random tap-error trials stayed within limits; this is not a reliable manual prescription.

Let readings change the plan. Our modeled controller checks two temperatures every minute, then reduces inflow only if the entire remaining plan passes for every compatible model. A unique mechanism need not be identified first. If no model explains the readings, stop relying on that certificate and reassess supply, motion and heat loss. Two specified shifts were detected after one minute; this is not a universal alarm or a tested emergency response. Two probes can also miss a cold corner outside the modeled conditions.

Count preparation as well as use. Across the broader model bank, cooling-only and hot-water trials lead to 26.09/23.63 L during a bath. For those fixed schedules, a reusable 6 L hot-water trial pays back from use 3 when conditions and reset costs match. Both trials need 30 minutes and a fresh uniform start. The online cases reduce the passive backup to 23.59/23.16 L under specified sensor errors; their repeated-use crossover with pulse is untested; trial/reset costs remain.

Keep the practical limits in view. An intact bubble layer may reduce heat loss, but the calculation does not measure a particular additive. Changed geometry, body position, motion, supply or foam needs reassessment. Use stored warmth, keep it distributed and avoid unnecessary overflow; the figures explain these tradeoffs rather than certifying bathing or hot-jet safety.

14 References and reproducible algorithm

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Algorithm and computational record

1. Form cell volumes and capacities; reject excessive body occupancy. Assemble symmetric face exchange, environmental/body conductances and the conservative stream. 2. Compute the mixed closed forms and energy lower bound. 3. Evaluate delayed constant-flow candidates with matrix exponentials and refine the first bracketed lower-temperature crossing. Reject ceiling or spread violations outside the jet zone. 4. Rank accepted candidates by added litres. 5. Independently integrate and check balances, analytic limits, parameter arithmetic and constraints; replay a finer mesh and perturb physical scenarios. 6. Optimize piecewise-constant and buffered schedules, then accept only candidates whose independent three-grid continuous bounds pass. 7. Rerun the constant-rate search at Sobol draws over the literature ranges. Independent RK45 checks use relative tolerance 2×10^{-9} , absolute $2 \times 10^{-10}^{\circ}\text{C}$ and steps of at most 5 s. AI assistance is acknowledged in [7].

Appendix A. Scenario definitions

Each row changes only the listed baseline inputs. All unspecified inputs remain as stated in Section 2; geometry and conductance are then rebuilt. The equal-volume aspect-ratio cases retain the envelope volume of 0.22425 m³. Widths s describe a smooth displacement distribution, not measured anatomical semiaxes.

Accepted candidates are independently replayed on their scenario network at intervals of no more than five seconds. These scenario checks are sampled checks; continuous-time envelopes are reported for the baseline constant rate and the selected scheduled policy; the latter uses additional interval refinement when necessary. “None accepted” refers to the documented finite search and is not an infeasibility theorem.

Table 14 lists the changes defining each scenario.

Table 14. Scenario definitions

Scenario	Changed input(s)
Weak Mixing	D (m ² /s) = 0.0003
Strong Mixing	D (m ² /s) = 0.003
Moving With Added Surface Loss	D (m ² /s) = 0.003; surface coefficient (W/m ² K) = 30
Loose Comfort	lower limit (°C) = 38
Tight Comfort	lower limit (°C) = 39.5
Foam	surface multiplier = 0.4
Shallow Wide	length (m) = 1.7; width (m) = 0.75; depth (m) = 0.175882
Deep Narrow	length (m) = 1.3; width (m) = 0.6; depth (m) = 0.2875
Small Bath	depth (m) = 0.2
Large Bath	depth (m) = 0.27
Deep Bath	depth (m) = 0.35
Larger Body	body displacement (m ³) = 0.075; body contact area (m ²) = 1.35
Long Body	body widths s (m) = [0.48, 0.12, 0.1]
Warmer Skin	skin temperature (°C) = 36
Cooler Skin	skin temperature (°C) = 32
High Loss	surface coefficient (W/m ² K) = 36; wall coefficient (W/m ² K) = 8.5; body coefficient (W/m ² K) = 40
Low Loss	surface coefficient (W/m ² K) = 17; wall coefficient (W/m ² K) = 4.5; body coefficient (W/m ² K) = 12
Stratified	vertical/horizontal D ratio = 0.2
Convective Surface Layer	vertical/horizontal D ratio = 3
Cool Supply	inlet temperature (°C) = 45

Report on Use of AI

Tool and scope

Codex (GPT-6-based) and Claude Sonnet 5.5 via Claude Code assisted modeling, calculation, validation, figures and writing on 7–10 October 2026; exact builds remain unverified. Post-sealing learning used full text of Outstanding papers 44845/54164, page overviews, selected figures and COMAP commentary; their data/figures were not copied. AI sub-agents supplied independent RHS/probe checks. This was not an unseen-problem trial.

Task

The task was to produce a complete historical MCM case; no conversation wording is reproduced.

Outputs, corrections and verification

Independent integration rejected the 19.35 L schedule and accepted a buffered alternative; fixed-policy replays then checked contact storage and flow paths. Baseline optimization was reused; coefficient/inlet-zone corrections remain recorded. Sources [1]–[4] and [6] were checked directly, and [10] at Theorems 3/6; [5]/[8] were not opened, so their relations were checked indirectly. AI produced the derivations and report from actual numerical results.

Control studies compared known-mixing schedules, pulse/passive ambiguity sets, alternative structures and failed candidates. New common-reserve candidates covered 5796 model-grid objects, with 54 independent extremal replays; two reserve failures were retained and corrected. Archived producers retain their versions. Analytical diffusion and inlet-cascade tests verify discretization. Baseline optimization was reused, and all observations are synthetic.

Finite-bank feedback gave nine complete services and 18 sampled replays; two shifts stopped at 660 s. Six predeclared off-bank truths gave two complete services and four refusals, with ten independent envelope replays. AI reviewed stored empty intervals, not missing historical predictions. An exact-remainder derivation then qualified narrow fixed-action parameter boxes, challenged by 102 independent RHS/energy replays; a failed loose bound was retained. AI reviewed the proof and numerical identities; no human verification or feedback-wide certificate is claimed.

Record limitations

The artifact retains the task description, tool identity, scope, code and numerical receipts, but lacks a full exported transcript and every intermediate AI output. These records do not substitute for human verification or physical bath experiments.